

Analyzing International Research Data in Stata Using the `repest` Package

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1 Introduction

The **repest** package in Stata is a module created by OECD analysts (*Avvisati, F. and F. Keslair (2014), REPEST: Stata module to run estimations with weighted replicate samples and plausible values, Statistical Software Components S457918, Boston College Department of Economics.*), to support analyses using plausible values (PVs) and complex surveys implemented in major international large-scale assessments, such as. **PISA**, **TIMSS**, **PIRLS**, and **PIAAC**.

These studies use complex sampling designs and advanced techniques for scaling results. Correct analysis requires incorporating **analytical weights** and **plausible values (PVs)** in the model to correct for measurement uncertainty and study design. In practice, this involves repeatedly performing analyses and averaging results. **repest** automates this process.

1.1 Why use the repest package?

Advantages of the repest package

- **Automation:** Simplifies the processing of complex data, automatically incorporating OECD and IEA survey designs such as replicate weights and plausible values (PV). This eliminates the need to manually account for weights and replications in analyses.

- **Convenience:** The package offers easily accessible functions for analyzing basic statistics (means, frequencies, correlations), statistical tests, and creating and exporting summaries.
- **Flexibility:** Enables analysis of data from multiple international educational studies.

1.1.1 Limitations of `repest`

⚠ Important limitations

Despite its advantages, `repest` has certain limitations. The package facilitates handling complex data, but it is not a comprehensive tool for all types of analyses. For **more advanced analyses** (e.g., multilevel models), other solutions will be more convenient – we mention them at the end of this guide.

The second limitation concerns data merging. `repest` assumes that the user already has a file ready to load. In practice, preparing such a file is one of the biggest challenges, especially in IEA studies (TIMSS, PIRLS, ICCS, ICILS). Although data can be merged directly in Stata, the process is complex and prone to errors. A better alternative is the free IEA IDB **Analyzer** program, which automates this workflow. We describe it in a separate guide. IEA IDB **Analyzer** makes it easier to select countries, merge different file types (student, school, and parent data), and restrict the dataset to the variables actually needed. The resulting smaller .sav (SPSS) file can then be loaded into Stata.

1.1.2 Supported studies

Study (Organization)	Supported
TIMSS (IEA)	
PIRLS (IEA)	
ICILS (IEA)	
ICCS (IEA)	
PIAAC (OECD)	
PISA (OECD)	
TALIS (OECD)	
SSES (OECD)	

1.2 Package installation

To get started, you need to install the **repest** package once. In the Stata command window, type:

```
ssc install repest, replace
```

2 Syntax description and repest options

In this section, we take a closer look at **repest** syntax and its options. Some of them are illustrated in the examples in later sections of the guide.

2.1 repest command syntax

The basic syntax of the **repest** command is as follows:

```
repest svname [if warunek] [in zakres] , estimate(cmd [,cmd_options]) [options]
```

- **svname**: Must be one of the study names supported by (e.g., PISA, TIMSS, PIRLS).
 - **estimate(cmd)**: ey option where you specify the type of analysis.
-

2.2 estimate() option

2.2.1 Built-in repest commands repest

- **means**: Calculates means.

```
repest PISA, estimate(means pv@math) by(cnt) over(gender) display
```

- **summarize**: Calculates descriptive statistics. Requires the **stats()** option.

```
repest PISA, estimate(summarize escs, stats(p5 p25 median p75 p95))
```

- **corr**: Calculates correlation matrix.

```
repest PISA, estimate(corr pv@math pv@read) by(cnt)
```

- **quantiletable**: Creates quantile tables similar to those in PISA reports.

```
repest PISA, estimate(quantiletable escs pv@math, nquantiles(5))
```

2.2.2 Stata commands

You can use any standard Stata command that accepts weights (**pweights** or **awweights**) by prefixing it with **stata**:

- **Linear regression (reg)**:

```
repest PISA, estimate(stata: reg pv@math escs i.gender)
```

- **Logistic regression (logit)**:

```
repest PIAAC, estimate(stata: logit zmienna_binarna pvnum@ age)
```

2.3 Key repest options

- **by(varname [, by_options])**: Performs analysis separately for each category of variable **varname** (e.g. **country**).
 - **levels(string)**: Restricts analysis to listed groups (e.g. **levels(POL FRA DEU)**).
 - **average(string)**: Calculates averaged results for pre-defined groups of countries (e.g. **average(OECD EU)**).
- **over(varlist [, test])**: Estimates statistics in subgroups within each country or overall.
- **outfile(filename, replace)**: Saves results to a stata file (**.dta**).
- **display**: Displays results in the console window.
- **store(name)**: Saves regression results to memory for later export via **esttab**.

- **results(results_options):** Controls which results are displayed.
 - **add(addlist):** Adds scalars (e.g. `r2`, `N`).
 - **combine(name: expression):** Estimates new statistics based on results.
- **fast:** Speeds up calculations for large data files (useful in preliminary analyses).
- **flag:** Replaces results based on a small number of observations with special missing code.
- **coverage:** Calculates the percentage of cases included in the analysis.
- **svyparms(svy_options):** Allows overriding default study parameters (required for `svyname = SVY`).

2.4 Best practices

Things to remember

- **Data verification:** Always use command `describe`, `summarize`, `tabulate`, `codebook`.
- **Script documentation:** Always document your scripts in `.do` or `.qmd`, files to ensure replicability.
- **Results verification:** Always compare obtained results with official international reports.
- **Handling of missing data:** Remember that `repest` by defaults omits observations with missing data.

2.5 Limitations and alternatives

`Repest` is great for generating descriptive statistics summaries from many countries. It has simple and logical syntax and supports many studies. It is also used by OECD and actively maintained by analysts associated with OECD.

- **pv :** Another package (`ssc install pv`). Works slower than `repest`, but it is more flexible. Allows manual definition of weight names, eliminating the needs to generate weights for TIMSS/PIRLS. It can be also combined with frequently used Stata commands (e.g. `estimates store`, `margins`, etc.).
- **mi i svyset:** Suitable for more advanced users. Its main advantage is the ability to use Stata's built-in commands for analyzing multiply imputed data.

2.6 Downloading data

Data from international studies are available on the following pages:

- **PISA:** www.oecd.org/pisa/data
- **PIAAC:** <https://www.oecd.org/skills/piaac/data/>
- **TALIS:** <https://www.oecd.org/en/about/programmes/talis.html#data>
- **SSES:** <https://www.oecd.org/en/about/programmes/oecd-survey-on-social-and-emotional-skills.html#data>
- **TIMSS, PIRLS, ICCS, ICILS:** <https://www.iea.nl/data-tools/repository>

For some OECD studies, you need to complete a short questionnaire before downloading the dataset.

2.7 Additional information about data

2.7.1 IEA data structure

IEA datasets (TIMSS, PIRLS, and ICILS) are usually divided into a large number of files, grouped by country, grade level, and research instrument used. After downloading TIMSS 2023 data for Grade 4, we will see over 500 files in the folder.

IEA file naming

Example: File `asapolm8.sav` contains data for Polish students from TIMSS 2023 for grade 4 in `.sav` format.

Where: - **asa**: survey answers and tasks results of 4th graders - **pol**: country code (Poland) - **m8**: study cycle (TIMSS 2023)

First letter in the file name indicates the grade level: - **a** – 4th grade - **b** – 8th grade

Next letters refer to the data type: - **asa/bsa** – students' results and plausible values (PVs) - **asp/bsp** – process data (e.g. reaction times) - **ash** – parent questionnaire data - **asg/bsg** – student questionnaire data - **acg/bcg** – school questionnaire data - **atg/btg** – teacher questionnaire data

2.7.2 PISA data structure

In PISA study data is structured differently than in IEA studies. Each country has its own file, and the data is divided into several files based on the type of survey tool (student, school, teacher) and the cycle of the study.

PISA file naming

Example: File CY08MSP_STU_QQQ.sav contains data from student questionnaires from all countries in the study conducted in 2022.

Where: - CY08 – study cycle (here: 2022) - MSP – Main Study - STU_QQQ – students' data

Other identifiers: - SCH_QQQ – school questionnaire - TCH_QQQ – teacher questionnaire - STU_COG – students' results in cognitive tests (reading, math, science etc.) - STU_FLT – results of tests of financial education - STU_ICT – results of tests of information and communication technology - STU_WBQ – questionnaire on students' well-being

2.8 Appending (adding observations from other datasets)

The `repest` package requires all data to be in one active dataset.

```
use "path/country_A.dta", clear
append using "path/country_B.dta"
append using "path/country_C.dta"
```

2.8.1 Merging data (adding variables from other datasets) (merge)

To merge student data with school data (m:1) or parent data(1:1):

```
* Load student data
use "path/student_data.dta", clear

* Merge school data (many students to one school)
merge m:1 schoolid using "path/school_data.dta"
tab _merge
drop _merge
```



```
* Merge parent data (one student to one parent)
merge 1:1 studid using "C:/path/parent_data.dta"
tab _merge
drop _merge
```

2.9 Understanding data structure

For **repest** to work correctly, variables in the dataset must have a specific name structure corresponding to conventions in individual studies.

2.9.1 Plausible values

Plausible values are a set of multiple imputations of latent traits (e.g., abilities) that allow obtaining unbiased measurement errors. When **repest** encounters @ symbol in a variable name (e.g. **pv@math**), it assumes that multiple imputations are available for this variable (e.g. **pv1math**, ..., **pv10math**, in case of PISA 2022).

- **PISA:**
 - mathematics **pv1math**, **pv2math**, ..., **pv10math**;
 - reading comprehension: **pv1scie**, **pv2scie**, ..., **pv10scie**
- **TIMSS**
 - mathematics: **BSMMAT01**, ..., **BSMMAT05** (plausible values of mathematics scores; each variable corresponds to one imputation of PV)
 - science: **BSMSCI01**, ..., **BSMSCI05** (plausible values of science scores)
- **ICCS** civic knowledge and skills
- **PIRLS**: **BSRRD01**, ..., **BSRRD05**
- **PIAAC:**
 - reading comprehension: **PVLIT1**, ..., **PVLIT10**
 - mathematical skills: **PVNUM1**, ..., **PVNUM10**
 - adaptive problem solving (in round II): **PVAPS1**, ..., **PVAPS10**

2.10 Analytical weights

The second important type of variables are **analytical weights**. `repest` requires, specific weight variables to be available in the active dataset. Their names are automatically set by `repest` when specifying the study name.

- **Final weights:** E.g. `w_fstuw` (for PISA students), `TOTWGTS` (for ICCS/ICILS), `WGT` (for PIRLS/TIMSS).
- **Replicate weights:** E.g. `w_fstr1` to `w_fstr80` (PISA), `SRWGT1` to `SRWGT75` (ICCS/ICILS), `JR1` to `JR150` (PIRLS/TIMSS).

For some studies, e.g. **TIMSS** and **PIRLS**, it is necessary to create weights. We describe how to do this in the practical section.

In some studies, different study schemes are distinguished depending on the type of respondents, e.g., in the **TALIS** study there are several options: `TALISSCH`, `TALISTCH`, `TALISEC_STAFF`, `TALISEC_LEADER`, which are used depending on which group one wants to analyze.

To analyze teacher or school data in ICCS/ICILS studies, you need to appropriately set the study name in the `repest` command:

- **Teacher data:** use `ICCS_T` or `ICILS_T` as `svname` in `repest` command. This will automatically use appropriate teacher weights.
- **Schools data:** use `ICCS_C` or `ICILS_C` as `svname`, to use school weights.

The third important piece of information is the variable describing the country, usually referring to ISO codes.

Case sensitivity in Stata

Stata is **case sensitive**. Variable names for PV should have a case consistent with the definition in `repest`. You can change them with the `rename` command, e.g.:

```
* Changes names of all pv*math variables to uppercase
rename pv*math, upper

* Changes names of all variables to lowercase
rename *, lower
```

3 Analyzing PISA 2022 data

For hands-on examples we'll start with the PISA study.

3.1 Downloading data

- Go to the OECD page with PISA 2022 data: <https://www.oecd.org/pisa/data/2022database/>
- After completing a short survey, you'll see a list of files in SAS and SPSS format. These are data from different study modules. We need the student survey data, which also include test scores.
- Stata has built-in capability to import SAS files (but without the provided label file). SPSS files (.sav) can be imported with `import spss` command, but this command struggles with very large datasets and doesn't handle longer variable and value labels well.

Create a folder on your computer for data, e.g., C:/pisa_data/.

C:/pisa_data/.

Download the international database. This is a compressed (zip) file containing data in SPSS format. The unzipped file is called **CY08MSP_STU_QQQ.SAV**. Save it in your working directory

We'll use the `import spss` command to load the file. The command is available from Stata 16 onward. If you're using an earlier version, install the `usespss` package instead.

3.2 Loading data in Stata

```
* Set the path to our working folder
* Remember to provide your own path here!
cd "C:/pisa_data/"

* Load PISA 2022 dataset in SPSS format.
import spss using CY08MSP_STU_QQQ.SAV, clear
```

The file is large: it contains 1278 variables for 613,744 students.

! Issue with importing large SPSS files in Stata

Unfortunately, Stata doesn't handle importing very large SPSS files (.sav) very well, even if one has enough RAM. The import may fail on some systems.

You can import data from a SAS file (*.sas7bdat), however **you'll lose value labels** (e.g. **gender** variable will contain values 1, 2, but without labels). OECD provides a *.SAS file, containing label definitions, but Stata doesn't support it.

In this case, we can use **R (package haven)**. This allows you to import the file and save it as .dta format with full metadata preserved.

If you only need a subset of the data, consider using the **IEA IDB Analyzer**. This tool supports files in various formats, but not in Stata one. Still, using **IEA IDB Analyzer** can be helpful in managing files – e.g., when we want to create a dataset from several countries or combine data from different instruments (e.g., student and teacher surveys).

After loading the file, let's check if we have the most important variables that **repest** uses. For simplicity, we'll ask for the first 3 PV and 5 replicate weights.

```
. describe PV1MATH-PV3MATH W_FSTUWT W_FSTURWT1- W_FSTURWT5 CNT
```

Variable name	Storage type	Display format	Value label	Variable label
PV1MATH	double	%10.0g		Plausible Value 1 in Mathematics
PV2MATH	double	%10.0g		Plausible Value 2 in Mathematics
PV3MATH	double	%10.0g		Plausible Value 3 in Mathematics
W_FSTUWT	double	%10.0g		FINAL TRIMMED NONRESPONSE ADJUSTED STUDENT RI
W_FSTURWT1	double	%10.0g		FINAL TRIMMED NONRESPONSE ADJUSTED STUDENT RI
W_FSTURWT2	double	%10.0g		FINAL TRIMMED NONRESPONSE ADJUSTED STUDENT RI
W_FSTURWT3	double	%10.0g		FINAL TRIMMED NONRESPONSE ADJUSTED STUDENT RI
W_FSTURWT4	double	%10.0g		FINAL TRIMMED NONRESPONSE ADJUSTED STUDENT RI
W_FSTURWT5	double	%10.0g		FINAL TRIMMED NONRESPONSE ADJUSTED STUDENT RI
CNT	str3	%7s		Country code 3-character

3.3 PISA Analyses Step by Step - Examples

3.3.1 Calculating Mean Score for Poland and Czech Republic

Let's limit our dataset to data for Poland and Czech Republic only. In Stata, we can use the **keep if** command (keep only) or the **drop if** command (delete if). In our case, we want to

keep only data where the country acronym (consistent with ISO) is POL (Poland) or CZE (Czech Republic).

```
. keep if CNT=="POL" | CNT=="CZE"  
(599,273 observations deleted)
```

3.4 Case sensitivity

In Stata, variable names are case-sensitive. A common problem with **repest** is inconsistent variable naming. For PISA data, **repest** expects key variables to be in lowercase. If **repest** cannot find a required variable, e.g., cnt, it will display an error message..

In our case, it is necessary to change the case of key variables defining the study. In this example, these are: CNT (country), CNTRYID (country id), CNTSCHID (school id), and CNTSTUID (student id).

...

```
rename CNT W_FSTUWT W_FSTURWT* CNTSCHID, lower
```

Now we can run the t analyses. Let's first look at some descriptive statistics for mathematics in Poland.

```
. repest PISA if cnt == "POL", estimate(means PV@MATH)
```

Survey parameters have been changed to PISA2015

_pooled.....

: _pooled

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PV_MATH_m	488.9601	2.27485	214.94	0.000	484.5014	493.4187

The results of this command show the estimated mean, its standard error, and 95% confidence interval.

Note that a comment **Survey parameters have been changed to PISA2015** also appears. This indicates that analyses are being conducted for 10pv (since PISA 2015, 10 pv are used in the study – in earlier editions 5 pv were used).

3.5 Gender differences reand significance test

Are there differences in mathematics scores between boys and girls?

To check this, we will use the `over()` option, which. This repeats the analysis for each group defined by the specified variable — in this case, gender: ST004D01T.

Adding the `test` option “tells” `repest` to calculate the difference estimate and test whether the difference between group means is statistically significant.

```
. repest PISA if cnt == "POL", estimate(means PV@MATH) over(ST004D01T, test) display
Survey parameters have been changed to PISA2015
over ST004D01T = 1 2

over var is ST004D01T

_pooled - ST004D01T = 1 .....
_pooled - ST004D01T = 2 .....
_pooled : _pooled
```

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ST004D01T=1							
PV_MATH_m		486.1626	2.946798	164.98	0.000	480.387	491.9382
ST004D01T=2							
PV_MATH_m		491.6791	2.672776	183.96	0.000	486.4405	496.9176
ST004D01T=d							
PV_MATH_m		5.51646	3.326107	1.66	0.097	-1.002589	12.03551

The results show that, the average mathematics results (PV_MATH_m) separately for boys (ST004D01T=1: 486.16) and girls (ST004D01T=2: 491.68). The row ST004D01T=d presents **the difference** between girls' and boys' (5.52 points). P>|z| is the p-value, which indicates statistical significance. In this case, the p-value (0.097) is above the chosen threshold of 0.05, which means that the difference in mean results **is not statistically significant**.

3.6 Differences in percentiles

When interpreting results, means alone don't tell the whole story. It's always worth looking at the variation in results. Let's look at selected percentiles for Poland.

```

.
. repest PISA if cnt == "POL", estimate(summarize PV@MATH, stats(p5 p25 p50 p75 p95))
Survey parameters have been changed to PISA2015

_pooled.....
: _pooled
-----+-----
          | Coefficient  Std. err.      z    P>|z|    [95% conf. interval]
-----+-----
PV_MATH_p5 |   340.3515   3.870307    87.94   0.000    332.7658   347.9372
PV_MATH_p25 |   426.3688   3.188472   133.72   0.000    420.1195   432.6181
PV_MATH_p50 |   490.4081   2.941234   166.74   0.000    484.6434   496.1728
PV_MATH_p75 |    551.97    2.612002   211.32   0.000    546.8506   557.0894
PV_MATH_p95 |   634.7251   3.701408   171.48   0.000    627.4705   641.9797
-----+-----

. repest PISA if cnt == "POL", estimate(summarize PV@MATH, stats(p10 p25 p50 p75 p90)) over(
Survey parameters have been changed to PISA2015
over ST004D01T = 1 2

over var is ST004D01T

_pooled - ST004D01T = 1 .....
_pooled - ST004D01T = 2 .....
_pooled : _pooled
-----+-----
          | Coefficient  Std. err.      z    P>|z|    [95% conf. interval]
-----+-----
ST004D01T=1 |
PV_MATH_p10 |   374.9559   4.83756    77.51   0.000    365.4745   384.4373
PV_MATH_p25 |   428.5824   3.732784   114.82   0.000    421.2663   435.8985
PV_MATH_p50 |   487.2313   3.528813   138.07   0.000    480.315    494.1476
PV_MATH_p75 |   544.6233   3.83937    141.85   0.000    537.0983   552.1483
PV_MATH_p90 |   593.7016   4.100419   144.79   0.000    585.6649   601.7383
-----+-----
ST004D01T=2 |
PV_MATH_p10 |   365.8842   4.476712    81.73   0.000    357.11    374.6584
PV_MATH_p25 |   423.8901   4.072185   104.09   0.000    415.9088   431.8714
PV_MATH_p50 |   494.1357   3.850228   128.34   0.000    486.5894   501.682
PV_MATH_p75 |   559.0357   3.361573   166.30   0.000    552.4471   565.6243
PV_MATH_p90 |   613.7861   4.009809   153.07   0.000    605.927    621.6452

```

ST004D01T=d							
PV_MATH_p10		-9.0717	6.414851	-1.41	0.157	-21.64458	3.501176
PV_MATH_p25		-4.6923	4.729754	-0.99	0.321	-13.96245	4.577848
PV_MATH_p50		6.9044	4.563641	1.51	0.130	-2.040172	15.84897
PV_MATH_p75		14.4124	4.640495	3.11	0.002	5.317198	23.5076
PV_MATH_p90		20.0845	5.090743	3.95	0.000	10.10683	30.06217

The table shows percentiles for each group (boys and girls). For example, PV_MATH_p10 for the 10th percentile, PV_MATH_p50 for the median and so on. Rows labeled ST004D01T=d show the differences in percentiles between girls and boys. Note the p-values: differences at lower percentiles (p10, p25) are **not statistically significant**, but at higher percentiles (p75, p90) they are. The p-values are 0.002 and 0.000 respectively.

Below are additional functions of the **repest** package for managing analysis results: displaying, saving, and reusing them:

- **display**: This is the default setting. **Repest** shows results directly in the Stata results window.
- **outfile("filename.dta", [suboptions])**: Saves results to a .dta file, facilitating its further processing. You can add suboptions: (**p-values**) or **se** (standard errors), to include more statistical details.
- **store(object_name)**: Saves results as objects in Stata that can be recalled later. A separate object is created for each level in **by()**.

3.7 How to compare results between countries and test for significant differences?

To compare results between countries, we use the **by** option.

```
* Analysis for Poland and Czech Republic with country breakdown

repest PISA, estimate(means PV@MATH) by(cnt, levels(POL CZE))

Survey parameters have been changed to PISA2015
```


POL.....

cnt : POL

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PV_MATH_m	488.9601	2.27485	214.94	0.000	484.5014	493.4187

CZE.....

cnt : CZE

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
PV_MATH_m	486.9992	2.093645	232.61	0.000	482.8957	491.1027

Is the mean in Poland higher than in Czech Republic in 2022?

* Analysis for Poland and Czech Republic by country

```
. repost PISA, estimate(means PV@MATH) over(cnt, test)
Survey parameters have been changed to PISA2015
option over() only allows numeric variables

over var is cnt
__00000U not found
r(111);
```

! Error option over() only allows numeric variables

In the above command, **repost** reported an error. The variable used in **over** cannot be a string variable. You should then use the second identifier available in the dataset, the **CNTRYID** variable, which is the numerical ISO country code (Poland is 616, and Czech Republic is 203).

```
. repost PISA, estimate(means PV@MATH) over(CNTRYID , test)
```

Survey parameters have been changed to PISA2015 over CNTRYID = 203 616

over var is CNTRYID

	_pooled - CNTRYID = 203		_pooled - CNTRYID = 616		_pooled : _pooled	
					Coefficient Std. err. z P> z	
	[95% conf. interval]		+-----+-----		=203	
PV_MATH_m	486.9992	2.093645	232.61	0.000	482.8957	491.1027
=616 PV_MATH_m	488.9601	2.27485	214.94	0.000	484.5014	493.4187
=d PV_MATH_m	1.960846	2.944474	0.67	0.505	-3.810218	7.731909

The average results of students from Poland and Czech Republic **do not differ significantly. point difference is not significant (p = 0.505). Since data come from a sample, we account for

```
::: {.callout-note}
```

```
## Differences between `by` and `over`
```

It's worth noting the differences between `by` and `over`.

The `by` prefix performs the analysis separately for each group, treating them as independent. However, the `over(\*variable\*, test)` option in `repest` analyzes all groups simultaneously v

```
:::
```

```
### Proportions of Students by Proficiency Levels
```

PISA reports often present data in the form of proficiency levels - usually as percentages of

While you could run the previous analyses by copying individual lines into the command window (file`). This is a text file (`.do`) containing a sequence of commands, functioning as a script

Do-files are essential for reproducibility, transparency, and efficient organization of any s

To calculate proportions of students, we need to transform PV (for mathematics these are var

```
```stata
```

```
* Define value labels for the new variable
```

```
label define math_level_labels 0 "Below 1c" ///
 1 "Level 1c" ///
 2 "Level 1b" ///
 3 "Level 1a" ///
 4 "Level 2" ///
 5 "Level 3" ///
 6 "Level 4" ///
 7 "Level 5" ///
 8 "Level 6"
```

```

foreach i of numlist 1/10 {

* Recode math PVs into proficiency levels
recode PV`i'MATH (min/233.169 = 0 "Below 1c") ///
 (233.17/295.469 = 1 "Level 1c") ///
 (295.47/357.769 = 2 "Level 1b") ///
 (357.77/420.069 = 3 "Level 1a") ///
 (420.07/482.379 = 4 "Level 2") ///
 (482.38/544.679 = 5 "Level 3") ///
 (544.68/606.989 = 6 "Level 4") ///
 (606.99/669.299 = 7 "Level 5") ///
 (669.30/max = 8 "Level 6"), gen(math`i'level)

* Apply value labels to the new variable
label values math`i'level math_level_labels

}

```

#### Explanation of the PV recoding script

Explanation of the PV recoding script

- **label define math\_level\_labels ...:** Defines descriptive labels (e.g., “Level 1c”, “Level 6”) for numeric values that will be assigned to levels. This ensures that results display readable names, instead of just numbers.
- **foreach i of numlist 1/10 { ... }:** This is a loop that performs the same command set ten times (for each of ten PVs, i.e., PV1MATH, PV2MATH, ..., PV10MATH). This prevents code repetition.
- **recode PViMATH (min/233.169 = 0 ...) ..., gen(mathilevel):** The recode command transforms each raw PISA score (PV1MATH, PV2MATH etc.) into a new variable (**math1level**, **math2level** etc.) by assigning a value (0 to 8) based on defined score bands (e.g., score from 233.17 to 295.469 becomes 1). The **gen()** option creates a new variable, keeping the original data intact.
- **label values mathilevel math\_level\_labels:** Finally, the pre-defined labels are applied to the new variable (**math1level**, **math2level**, etc.), so that the numeric values (0, 1, 2, etc.) are displayed as meaningful level descriptions (like “Below 1c”, “Level 1c”).

The above commands recode PV values into all levels, but we can also simplify, e.g., by distinguishing only students below level 2 (value 1) and those at level 2 or above (value 0).

With these variables, we can calculate the percentages of students at different levels. Let’s

compute this for Poland.

```
. repeat PISA if cnt=="POL" , estimate(freq math@level)
Survey parameters have been changed to PISA2015
```

```
_pooled.....
: _pooled
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
math_level_0	.0740546	.0629373	1.18	0.239	-.0493001	.1974094
math_level_1	1.129739	.2436488	4.64	0.000	.6521962	1.607282
math_level_2	6.367772	.5078255	12.54	0.000	5.372453	7.363092
math_level_3	15.4304	.7576769	20.37	0.000	13.94538	16.91542
math_level_4	23.75994	.9219499	25.77	0.000	21.95295	25.56692
math_level_5	25.58255	.8678128	29.48	0.000	23.88167	27.28344
math_level_6	18.24058	.6846419	26.64	0.000	16.89871	19.58245
math_level_7	7.499555	.4674799	16.04	0.000	6.583311	8.415799
math_level_8	1.915414	.2991674	6.40	0.000	1.329057	2.501771

### 3.7.1 Linear regression

In this example, we'll analyze the association between students' socio-economic status (ESCS) and their math scores (PV@MATH).

The **results(add(r2 N))** option: This option allows you to add extra statistics to the results table, which Stata stores by default. Here, we add **R-squared** (e\_r2) - a measure of model fit, and **number of observations** (e\_N).

We'll conduct the analyses separately for Poland and the Czech Republic.

```
* Linear regression model

* Standard, linear effect

. repeat PISA, estimate(stata: reg PV@MATH ESCS) by(cnt) results(add(r2 N))
```

Survey parameters have been changed to PISA2015

`"CZE"' ` "POL"'

CZE.....

cnt : CZE

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	ESCS	50.58072	1.800561	28.09	0.000	47.05168	54.10975
	_cons	492.6841	1.810705	272.10	0.000	489.1352	496.2331
	e_r2	.2204457	.0121808	18.10	0.000	.1965719	.2443196
	e_N	8301	.	.	.	.	.

POL.....

cnt : POL

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	ESCS	40.34929	1.857624	21.72	0.000	36.70842	43.99017
	_cons	494.8693	1.851499	267.28	0.000	491.2404	498.4981
	e_r2	.1625078	.0129805	12.52	0.000	.1370666	.187949
	e_N	5875	.	.	.	.	.

\* Nonlinear ESCS effect

. repeat PISA, estimate(stata: reg PV@MATH c.ESCS##c.ESCS) by(cnt) results(add(r2 N))

Survey parameters have been changed to PISA2015

`"CZE"' ` "POL"'

CZE.....

cnt : CZE

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	ESCS	49.8014	1.80252	27.63	0.000	46.26852	53.33427
c_ESCS	c_ESCS	-5.542008	1.618431	-3.42	0.001	-8.714074	-2.369941
	_cons	496.8342	2.284468	217.48	0.000	492.3567	501.3117
	e_r2	.2233969	.0118917	18.79	0.000	.2000896	.2467041
	e_N	8301	.	.	.	.	.

POL.....

cnt : POL

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	+						
	ESCS	40.10634	1.979861	20.26	0.000	36.22588	43.98679
c_ESCS_c	_ESCS	-1.108514	2.206676	-0.50	0.615	-5.433519	3.216491
	_cons	495.7264	2.62605	188.77	0.000	490.5794	500.8733
	e_r2	.1627424	.0128378	12.68	0.000	.1375808	.187904
	e_N	5875	.	.	.	.	.

\* Linear regression model

\* Standard, linear effect

. repeat PISA, estimate(stata: reg PV@MATH ESCS) by(cnt) results(add(r2 N))

Survey parameters have been changed to PISA2015

`"CZE"'`"POL"'

CZE.....

cnt : CZE

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	+						
	ESCS	50.58072	1.800561	28.09	0.000	47.05168	54.10975
	_cons	492.6841	1.810705	272.10	0.000	489.1352	496.2331
	e_r2	.2204457	.0121808	18.10	0.000	.1965719	.2443196
	e_N	8301	.	.	.	.	.

POL.....

cnt : POL

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	+						
	ESCS	40.34929	1.857624	21.72	0.000	36.70842	43.99017
	_cons	494.8693	1.851499	267.28	0.000	491.2404	498.4981
	e_r2	.1625078	.0129805	12.52	0.000	.1370666	.187949
	e_N	5875	.	.	.	.	.

- \* Nonlinear ESCS effect

```
. repost PISA, estimate(stata: reg PV@MATH c.ESCS##c.ESCS) by(cnt) results(add(r2 N))
Survey parameters have been changed to PISA2015
```

```
`"CZE" '  `'"POL" '
```

CZE . . . . .

cnt : CZE

		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	ESCS	49.8014	1.80252	27.63	0.000	46.26852	53.33427
c_ESCS_c_ESCS		-5.542008	1.618431	-3.42	0.001	-8.714074	-2.369941
	_cons	496.8342	2.284468	217.48	0.000	492.3567	501.3117
	e_r2	.2233969	.0118917	18.79	0.000	.2000896	.2467041
	e_N	8301	.	.	.	.	.

POL.....

cnt : POL

Model 1: c_ESCS_c_ESCS							
		Coefficient	Std. err.	z	P> z	[95% conf. interval]	
c_ESCS_c_ESCS	_cons	40.10634	1.979861	20.26	0.000	36.22588	43.98679
	e_r2	-.108514	2.206676	-.050	0.615	-5.433519	3.216491
	e_N	495.7264	2.62605	188.77	0.000	490.5794	500.8733
		.1627424	.0128378	12.68	0.000	.1375808	.187904
		5875	.	.	.	.	.

## i Interpretation of linear regression results

Results of the first regression model show a **linear relationship** between ESCS and math results in both countries.

- **For the Czech Republic:** An increase in ESCS by one unit is associated with an increase in math score of about **50.58 points**. The model explains about **22%** of the variance ( $e\_r2 = 0.22$ ).
- **For Poland:** The same ESCS increase results in about **40.35 points**, and the model explains about **16%** of variance ( $e\_r2 = 0.16$ ).

In both cases, the ESCS effect is statistically significant (p-values near 0.000). The

association between socio-economic status and educational results is stronger in the Czech Republic than in Poland.

#### **i** Nonlinear ESCS effect (self-interaction of variable)

In Stata, to model **nonlinear effects** for continuous variables (like ESCS), powers of that variable can be added to the model. Most often, the square of the variable is included. Stata facilitates this with **factor variable notation** `c.` denotes continuous variables, and `i.` categorical ones (like gender). The fragment `c.variable##c.variable` creates an interaction of a continuous variable with itself, which is an equivalent to including a quadratic term (`variable^2`). Stata will automatically create both the main effect (ESCS) and the square effect (`c.ESCS#c.ESCS`, often displayed as `c.ESCS_c_ESCS`). This is a convenient way to check if the effect is strictly linear.

In our example, the quadratic coefficient is **statistically significant in the Czech Republic** (p-value = 0.001), indicating that the association between ESCS and math results is nonlinear. The negative quadratic coefficient suggests that ESCS's positive impact on math results is stronger at lower ESCS levels and gradually weakens at higher status. R-squared increases slightly to 0.2234, suggesting the nonlinear model is a better fit to the data.

---

## 4 Analysis of TIMSS (and PIRLS) data

We'll now move on to the TIMSS survey. The main difference as compared to PISA is the necessity to **compute replicate weights manually**. Similar procedures apply to the PIRLS 2021 survey, so the scripts below will also work for PIRLS data.

### 4.1 File structure

Recall that IEA studies (TIMSS, PIRLS) often divide datasets into multiple files. A typical filename structure is as follows: **asa pol m8 .sav**

Where:

- **asa**: Data type, e.g., **asa** - 4th grade student data (for 8th grade: **bsa**).
- **pol**: Country code (Poland).
- **m8**: Survey cycle (e.g., TIMSS 2019).



Przypomnijmy, że w badaniach IEA (TIMSS, PIRLS) zbiory są często podzielone na wiele plików. Typowa struktura nazwy pliku wygląda następująco: **asa pol m8 .sav**

Gdzie:

- **asa**: Typ danych, np. **asa** - dane uczniów Klasy 4 (dla Klasy 8: **bsa**).
- **pol**: Kod kraju (Polska).
- **m8**: Cykl badania (np. TIMSS 2019).

---

## 4.2 Preparing TIMSS data

### 4.2.1 Downloading and combining data

TIMSS data are available from the IEA repository: <https://www.iea.nl/data-tools/repository>. Choose the TIMSS section, accept the terms, and you can download the data. TIMSS is conducted in grades 4 and 8. We're interested in grade 4. Data can be downloaded in three formats: SAS, SPSS, and R. Choose **SPSS**.

After downloading, copy the files to your working directory, The downloaded package contains multiple data files and documentation in separate folders. This is is very helpful.

Let's see what the directory looks like from Stata's window:

```
cd C:\timss_data\TIMSS2023_IDB_SPSS_G4
C:\timss_data\TIMSS2023_IDB_SPSS_G4

. dir
<dir> 6/14/25 13:48 .
<dir> 6/14/25 13:48 ..
<dir> 6/14/25 13:48 1_User Guide
<dir> 6/14/25 13:48 2_Data Files
<dir> 6/14/25 13:48 3_Supplemental Material
<dir> 6/14/25 13:48 4_Reports and Encyclopedia
0.1k 6/14/25 13:48 timss2023.org.url
```

We're interested in the Polish data. You'll find it in the "2\_Data Files" folder, with a subfolder "SPSS Data".

```
. cd "C:\timss_data\TIMSS2023_IDB_SPSS_G4\2_Data Files\SPSS Data\"
C:\timss_data\TIMSS2023_IDB_SPSS_G4\2_Data Files\SPSS Data

. dir *pol*
 67.3k 6/14/25 13:48 acgpolm8.sav
8426.9k 6/14/25 13:48 asapolm8.sav
5005.8k 6/14/25 13:48 asgpolm8.sav
1049.6k 6/14/25 13:48 ashpolm8.sav
 12.0M 6/14/25 13:48 asppolm8.sav
3152.6k 6/14/25 13:48 asrpolm8.sav
7076.1k 6/14/25 13:48 astpolm8.sav
 283.6k 6/14/25 13:48 atgpolm8.sav
```

We're interested in the datasets starting with **asa**, which we'll load and save in Stata format. In this example, we only use Polish data, but it's worth showing how to convert data for other countries as well.

```
usespss asapolm8.sav
save asapolm8.dta, replace
}
```

### 4.3 Preparing weights for repeat (Essential step!)

Unlike PISA, IEA studies (TIMSS/PIRLS) use the Jackknife replication method. Here, “sampling zones” (JKZONE variable) are crucial, because they determine how many replicate weights are generated for proper analysis.

IEA uses **Jackknife Repeated Replication** instead of **Balanced Repeated Replication** because samples in TIMSS and PIRLS have more varied structures across countries. In many education systems, it is not possible to obtain symmetrical, balanced subsamples required by BRR. JRR is simpler and more flexible - each country gets as many replicate weights as actual sampling zones.

Compared to previous editions, TIMSS 2023 increased the number of zones (from 75 to 125), but the number of zones differs by country. At the time of writing this guide, this is not yet reflected in the **repest** script.

First, it's crucial to use a flexible script for the data structure. The following script must be run only once after loading and joining the files.

```

* -----
* Generating weights for data with different JKZONE
* -----

* Main weight expected by repest
gen double WGT = TOTWGT

* Renumber zones 1..k (when JKZONE isn't 1..k)
tempvar z_zone
egen long `z_zone' = group(JKZONE)

* Number of zones k and number of replicates r = 2*k
quietly levelsof `z_zone', local(zs)
local k : word count `zs'
local r = 2*`k'

* JRR-paired replicate weights - exactly 2*k
capture drop JR*
forvalues i = 1/`k' {
 local j = `k' + `i'
 gen double JR`i' = WGT
 replace JR`i' = 2*WGT if `z_zone'==`i' & JKREP==1
 replace JR`i' = 0 if `z_zone'==`i' & JKREP==0

 gen double JR`j' = WGT
 replace JR`j' = 2*WGT if `z_zone'==`i' & JKREP==0
 replace JR`j' = 0 if `z_zone'==`i' & JKREP==1
}

* How many replicates? - Enter THIS NUMBER manually in NREP()
di as res "Zones (k): " `k'
di as res "Replicates (r=2*k): " `r'
ds JR*, has(type numeric)

```

This script creates the correct number of replicate weights (e.g., 150 for TIMSS 2019, or an appropriate number for TIMSS 2023) using Jackknife Repeated Replication.

The inconvenience is that analyses can only be conducted for a specific country, because repest requires precise specification of the number of replicates (omitting this produces incorrect standard errors). The end of the above script displays this information.

```

* How many replicates? - Enter THIS NUMBER manually in NREP()
di as res "Zones (k): " `k'
Zones (k): 76

di as res "Replicates (r=2*k): " `r'
Replicates (r=2*k): 152

```

We can see, that for Poland, there are 76 zones, and the script above created 152 replicate weights.

#### **i** Note

If you want to analyze teacher survey data, compute weights in the same way. Load the teacher file and generate weights using teacher weights `gen double WGT = MATWGT`. Proceed identically (JKZONE/JKREP variables are in the teacher data file).

## 4.4 TIMSS step-by-step analysis – examples

The basic commands are very similar to those shown above for PISA data analysis. Differences come down to variable names and the number of replicate weights.

Let's first compute mean results.

```

. . repest TIMSS , estimate(means ASMMAT0@) svyparms(NREP(152))

_pooled.....
: _pooled

 | Coefficient Std. err. z P>|z| [95% conf. interval]
-----+-----
ASMMAT0__m | 546.0228 2.033083 268.57 0.000 542.038 550.0076

```

### 4.4.1 Gender differences in Poland

Now, let's compute the differences between boys and girls in Poland. Computing mean differences is simple and does not differ from previous examples. We'll use `estimate(means`

ASMMAT00 ) over(ITSEX , test), substituting the IDCNTY variable for the gender variable ITSEX.

Below, we examine estimates for selected percentiles and differences between boys and girls.

```
repest TIMSS , estimate(summarize ASMMAT00, stats(mean p10 p25 p50 p75 p90)) over(ITSEX, test)
over ITSEX = 1 2
(file C:\Users\IBE-2660\AppData\Local\Temp\ST_2150_000005.tmp not found)
file C:\Users\IBE-2660\AppData\Local\Temp\ST_2150_000005.tmp saved
over var is ITSEX
```

```
_pooled - ITSEX = 1
_pooled - ITSEX = 2
_pooled : _pooled
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----						
ITSEX=1						
ASMMAT0__mean	540.6729	2.446792	220.97	0.000	535.8772	545.4685
ASMMAT0__p10	442.7281	3.944585	112.24	0.000	434.9968	450.4593
ASMMAT0__p25	493.9585	3.532112	139.85	0.000	487.0357	500.8814
ASMMAT0__p50	545.2986	3.510619	155.33	0.000	538.4179	552.1793
ASMMAT0__p75	591.5121	3.101972	190.69	0.000	585.4324	597.5919
ASMMAT0__p90	632.209	5.778426	109.41	0.000	620.8835	643.5345
-----+-----						
ITSEX=2						
ASMMAT0__mean	551.32	2.724234	202.38	0.000	545.9806	556.6594
ASMMAT0__p10	447.9956	6.53938	68.51	0.000	435.1787	460.8126
ASMMAT0__p25	503.0338	3.998141	125.82	0.000	495.1976	510.87
ASMMAT0__p50	557.1689	4.501254	123.78	0.000	548.3466	565.9912
ASMMAT0__p75	604.0593	3.711391	162.76	0.000	596.7851	611.3335
ASMMAT0__p90	646.2404	4.36668	147.99	0.000	637.6819	654.7989
-----+-----						
ITSEX=d						
ASMMAT0__mean	10.64712	3.188914	3.34	0.001	4.396965	16.89728
ASMMAT0__p10	5.267572	7.480926	0.70	0.481	-9.394773	19.92992
ASMMAT0__p25	9.075292	5.415111	1.68	0.094	-1.538131	19.68872
ASMMAT0__p50	11.87034	5.404963	2.20	0.028	1.276807	22.46387
ASMMAT0__p75	12.54714	4.43619	2.83	0.005	3.852365	21.24191
ASMMAT0__p90	14.03136	6.960439	2.02	0.044	.3891518	27.67357

The table shows **math percentile scores for girls (ITSEX=1) and boys (ITSEX=2) in**

## Poland and their differences (ITSEX=d).

The mean difference is 10.6 points and it is statistically significant, with a standard error of 3.2 points, and the 95% confidence interval for this difference is 4.4 to 16.9 points.

Percentile values suggest that boys' advantage is more pronounced at the top of the distribution. For example, the boys' median (p50) is higher by about 12 points and this difference is statistically significant. In the lower part of the distribution (p10, p25), differences are smaller and often not significant.

---

### 4.4.2 Proficiency levels

The TIMSS dataset contains a variable describing proficiency levels (actually five separate variables calculated for each `pv`). Using the `fre` command (`ssc install fre`), we inspect the first variable in this group:

```
. fre ASMIBM01
```

```
ASMIBM01 -- INTERN. MATH BENCH REACHED WITH 1ST PV
```

			Freq.	Percent	Valid	Cum.
		+				
Valid	1 Below 400		170	3.64	3.64	3.64
	2 At or above 400 but below 475		622	13.33	13.33	16.97
	3 At or above 475 but below 550		1509	32.34	32.34	49.31
	4 At or above 550 but below 625		1695	36.33	36.33	85.64
	5 At or above 625		670	14.36	14.36	100.00
	Total		4666	100.00	100.00	

This variable is a recoded `pv` variable, based on the study's defined level cutoffs. The above distribution is unweighted. To get a population distribution, we must use weights and average across all `pv` values. Here, `repest` helps:

```
. repest TIMSS , estimate(freq ASMIBM0@) svyparms(NREP(152))
```

```
_pooled.....
: _pooled
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ASMIBMO__1	3.818529	.453191	8.43	0.000	2.930291	4.706767
ASMIBMO__2	13.21859	.7538217	17.54	0.000	11.74113	14.69606
ASMIBMO__3	32.43265	.9518056	34.07	0.000	30.56714	34.29815
ASMIBMO__4	36.15421	1.047753	34.51	0.000	34.10065	38.20777
ASMIBMO__5	14.37602	.8519447	16.87	0.000	12.70624	16.0458

The **Coefficient** column gives the percent of students at each level; for example, the first row is for those scoring below 400 points on the TIMSS scale. In Poland, that's about **4% of students** (95% CI: 2.9–4.7%).

You can also compute more detailed comparisons. Let's return to the example with gender differences:

Możemy też wyliczyć bardziej rozbudowane porównania. Wróćmy do przykładu z różnicami ze względu na płeć:

```
. repeat TIMSS , estimate(freq ASMIBMO@) over(ITSEX, test) svyparms(NREP(152))
over ITSEX = 1 2
```

```
_pooled - ITSEX = 1
_pooled - ITSEX = 2
_pooled : _pooled
```

	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ITSEX=1						
ASMIBMO__1	3.807798	.6125007	6.22	0.000	2.607318	5.008277
ASMIBMO__2	14.41528	.9742022	14.80	0.000	12.50588	16.32468
ASMIBMO__3	34.56772	1.225612	28.20	0.000	32.16556	36.96988
ASMIBMO__4	35.13991	1.570727	22.37	0.000	32.06135	38.21848
ASMIBMO__5	12.06929	1.349484	8.94	0.000	9.424351	14.71423
ITSEX=2						
ASMIBMO__1	3.829155	.5573351	6.87	0.000	2.736798	4.921511
ASMIBMO__2	12.0337	1.033463	11.64	0.000	10.00815	14.05925
ASMIBMO__3	30.31861	1.280613	23.68	0.000	27.80865	32.82856
ASMIBMO__4	37.15851	1.629268	22.81	0.000	33.9652	40.35182
ASMIBMO__5	16.66003	1.167054	14.28	0.000	14.37265	18.94741

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ITSEX=d							
ASMIBMO__1		.0213571	.7412523	0.03	0.977	-1.431471	1.474185
ASMIBMO__2		-2.381582	1.325725	-1.80	0.072	-4.979955	.2167903
ASMIBMO__3		-4.24911	1.629858	-2.61	0.009	-7.443573	-1.054647
ASMIBMO__4		2.018596	2.418225	0.83	0.404	-2.721038	6.75823
ASMIBMO__5		4.590739	1.858332	2.47	0.013	.9484761	8.233002
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Comparing the percentages of students at different proficiency levels shows that boys more often attain the highest level (625+) and less often the lower levels. For example, 16.7% of boys reach the highest level, compared to 12.1% of girls. This difference is statistically significant.

The largest differences appear at levels 3 and 5—boys are less often at level 3 (30.3% vs 34.6%) and more often at the highest level (16.7% vs 12.1%). This suggests that there are more high-achieving boys than girls, although differences in means were moderate.